

Supplementary Appendix

Cost-Effectiveness of Intismeran Autogene (mRNA-4157/V940) plus Pembrolizumab versus Pembrolizumab Alone as Adjuvant Therapy for Resected Stage IIIB–IV Melanoma

Yichao Jin, Ph.D.

Contents

1	Model Structure and Assumptions	3
1.1	Health-State Definitions	3
1.2	State Occupancy Equations	3
1.3	Model Schematic	4
2	Survival Data and Parametric Extrapolation	4
2.1	Extracted Survival Probabilities	4
3	Clinical Input Provenance	5
3.1	Observed vs. Fitted Survival Curves	7
4	Survival Model Selection	8
4.1	Component-Level Parametric Fit	8
4.2	Joint Model Selection	8
4.3	Lifetime OS Extrapolation by Distribution	9
5	General-Population Mortality and Structural Validation	9
5.1	GP Mortality Implementation	9
5.2	Long-Term Survival Diagnostics	10
5.3	OS versus General-Population Survival	10
5.4	Structural Quality Assurance	11
6	External Plausibility Assessment	11
6.1	External Plausibility Landmarks	11
6.2	Longitudinal External Plausibility Assessment	12
7	Model Inputs	12
7.1	Complete Parameter Table	12
8	Additional Results	14
8.1	Lifetime Cost Decomposition	14
8.2	LY and QALY Decomposition	14
9	Uncertainty and Structural Scenario Analyses	14
9.1	Deterministic Sensitivity Analysis	14
9.2	Probabilistic Sensitivity Analysis	15
9.2.1	PSA Implementation Details	15
9.2.2	PSA Results	17
9.2.3	PSA Convergence	17

9.2.4	Cost-Effectiveness Plane	18
9.3	Structural Scenario Analyses	18
10	Deterministic Threshold Acquisition Price	20
10.1	Price Threshold Analysis	20
10.2	ICER versus Acquisition Price	21
11	Reproducibility and Reporting	21
11.1	Technical Reproducibility	21
11.2	CHEERS 2022 Checklist	22

1 Model Structure and Assumptions

1.1 Health-State Definitions

Table S1: Health-State Definitions, Clinical Interpretation, and Model Inputs

State	Clinical Definition	Utility	Monthly Cost
RF	Recurrence-free after complete resection; receiving adjuvant pembrolizumab (up to 18 cycles) with or without ipilimumab (up to 9 doses)	0.83	\$0
LR	Confirmed locoregional recurrence without distant metastasis; includes salvage surgery, re-treatment, and surveillance	0.64	\$3,000
DM	Confirmed stage IV melanoma with distant metastasis; includes systemic therapy, palliative care, and monitoring	0.55	\$12,000
Death	All-cause mortality	0	\$0

Note: Utilities and costs are drawn from published melanoma CEA literature [1–3]. Costs are in 2026 USD. See Supplementary Table S13 for full PSA distribution parameters.

1.2 State Occupancy Equations

State occupancy at each monthly cycle is determined by the partitioned survival model (PSM) approach, using three parametric survival functions:

- **RFS(t)** — Recurrence-free survival: probability of remaining alive and free of any recurrence (locoregional or distant) at time t .
- **DMFS(t)** — Distant metastasis-free survival: probability of remaining alive and free of distant metastasis at time t .
- **OS(t)** — Overall survival: probability of being alive at time t .

The occupancy equations are derived from the ordering of these three events:

- All patients who are recurrence-free are also distant metastasis-free and alive: $RF = RFS$.
- Patients who have had a locoregional recurrence but have not yet developed distant metastasis are those who are distant metastasis-free minus those who are recurrence-free: $LR = DMFS - RFS$.
- Patients who are alive with distant metastasis are those who are alive minus those who are distant metastasis-free: $DM = OS - DMFS$.
- The dead state is the complement of overall survival: $Dead = 1 - OS$.

The constraint $RFS \leq DMFS \leq OS$ is enforced at all time points to ensure non-negative state occupancy. This ordering reflects the endpoint definitions used in the clinical trials: DMFS counts either distant metastasis *or death* as an event, so the probability of remaining alive and distant-metastasis-free cannot exceed overall survival ($DMFS \leq OS$). Similarly, RFS counts recurrence *or death*, so $RFS \leq DMFS$. The ratio-constrained structure $r_{LR} = RFS/DMFS$ and $r_{DM} = DMFS/OS$ guarantees this ordering by construction.

1.3 Model Schematic



Figure S1: Partitioned survival model structure. The model uses four mutually exclusive health states. State occupancy is derived from three parametric survival functions (RFS, DMFS, OS) rather than from transition probabilities. Arrows denote the ratio-constrained relationship $RFS \leq DMFS \leq OS$.

2 Survival Data and Parametric Extrapolation

2.1 Extracted Survival Probabilities

Survival probabilities were extracted from the published Kaplan–Meier curves (Figure 1) of Khattak et al. (2026) [4] using WebPlotDigitizer v4.7. Data were extracted at 18, 24, 36, 48, and 60 months for OS and RFS, and at 18, 24, 36, and 48 months for DMFS. The extracted values are reported in Table S2.

Table S2: Extracted KEYNOTE-942 Survival Probabilities (%)

Month	RFS (%)		DMFS (%)		OS (%)	
	Combo	Pembro	Combo	Pembro	Combo	Pembro
18	80.4	62.2	92.0	73.0	96.0	93.4
24	78.3	60.0	90.8	70.6	96.0	93.4
36	73.8	55.6	85.6	65.4	93.8	85.6
48	72.4	49.1	83.9	65.4	92.2	85.6
60	68.8	49.1	—	—	92.2	71.3

Digitization method: WebPlotDigitizer v4.7, manual extraction. The 60-month landmark values match the published 5-year ASCO 2026 update [4]. At earlier time points, the 18-month RFS rates from the primary analysis [5] differ from digitized values by 1.4 pp (combo) and 0.2 pp (pembro), consistent with typical digitization error. The 5-year RFS 95% CIs: combo 56.3–78.3%, pembro 33.3–63.0%.

3 Clinical Input Provenance

Table S3 summarizes the provenance of all clinical inputs used in the model. The evidence status column classifies each input into one of five categories:

- **Published:** The value is directly reported in the cited source.
- **Digitized:** The value was extracted from a published Kaplan–Meier curve using Web-PlotDigitizer v4.7.
- **Derived:** The value is mathematically derived from other inputs (e.g., ratio of survival functions).
- **Modeled:** The value is produced by the model’s parametric extrapolation (e.g., long-term survival beyond observed data).
- **Topline only:** The source reports a statistically significant result but the numerical hazard ratio or landmark survival probabilities have not yet been disclosed.

Table S3: Clinical Input Provenance

Input	Value	Evidence Status	Source
Input	Value	Evidence Status	Source
<i>KEYNOTE-942 (Phase 2b, 5-year follow-up)</i>			
Recurrence-Free Survival (RFS)			
18-month RFS, combo	80.4%	Digitized	Fig. 1, Khattak 2026
24-month RFS, combo	78.3%	Digitized	Fig. 1, Khattak 2026
36-month RFS, combo	73.8%	Digitized	Fig. 1, Khattak 2026
48-month RFS, combo	72.4%	Digitized	Fig. 1, Khattak 2026
60-month RFS, combo	68.8%	Published	Khattak 2026
60-month RFS, combo 95% CI	56.3–78.3%	Published	Khattak 2026
18-month RFS, pembro	62.2%	Digitized	Fig. 1, Khattak 2026
24-month RFS, pembro	60.0%	Digitized	Fig. 1, Khattak 2026
36-month RFS, pembro	55.6%	Digitized	Fig. 1, Khattak 2026
48-month RFS, pembro	49.1%	Digitized	Fig. 1, Khattak 2026
60-month RFS, pembro	49.1%	Published	Khattak 2026
60-month RFS, pembro 95% CI	33.3–63.0%	Published	Khattak 2026
Distant Metastasis-Free Survival (DMFS)			
18-month DMFS, combo	92.0%	Digitized	Fig. 1, Khattak 2026
24-month DMFS, combo	90.8%	Digitized	Fig. 1, Khattak 2026
36-month DMFS, combo	85.6%	Digitized	Fig. 1, Khattak 2026
48-month DMFS, combo	83.9%	Digitized	Fig. 1, Khattak 2026
60-month DMFS, combo	–	Not reported	N/A
18-month DMFS, pembro	73.0%	Digitized	Fig. 1, Khattak 2026
24-month DMFS, pembro	70.6%	Digitized	Fig. 1, Khattak 2026
36-month DMFS, pembro	65.4%	Digitized	Fig. 1, Khattak 2026
48-month DMFS, pembro	65.4%	Digitized	Fig. 1, Khattak 2026
60-month DMFS, pembro	–	Not reported	N/A
Overall Survival (OS)			

Input	Value	Evidence Status	Source
18-month OS, combo	96.0%	Digitized	Fig. 1, Khattak 2026
24-month OS, combo	96.0%	Digitized	Fig. 1, Khattak 2026
36-month OS, combo	93.8%	Digitized	Fig. 1, Khattak 2026
48-month OS, combo	92.2%	Published	Khattak 2026
60-month OS, combo	92.2%	Published	Khattak 2026
18-month OS, pembro	93.4%	Digitized	Fig. 1, Khattak 2026
24-month OS, pembro	93.4%	Digitized	Fig. 1, Khattak 2026
36-month OS, pembro	85.6%	Digitized	Fig. 1, Khattak 2026
48-month OS, pembro	85.6%	Published	Khattak 2026
60-month OS, pembro	71.3%	Published	Khattak 2026
Hazard Ratios			
RFS HR (combo vs. pembro)	0.510	Published	Khattak 2026
RFS HR 95% CI	0.294–0.887	Published	Khattak 2026
DMFS HR (combo vs. pembro)	0.411	Published	Khattak 2026
DMFS HR 95% CI	0.200–0.843	Published	Khattak 2026
OS HR (combo vs. pembro)	0.471	Published	Khattak 2026
OS HR 95% CI	0.165–1.345	Published	Khattak 2026
OS events	14 (7 per arm)	Published	Khattak 2026
<i>INTerpath-001 (Phase 3, topline announcement)</i>			
RFS	Statistically significant	Topline only	Merck/Moderna 2026
DMFS	Statistically significant	Topline only	Merck/Moderna 2026
Numerical HR / KM curves	Not yet disclosed	Topline only	N/A
<i>Modeled / Derived Values</i>			
Fitted log-normal parameters			
OS μ (combo), σ (combo)	8.396, 3.096	Derived	Least-squares fit
OS μ (pembro), σ (pembro)	4.841, 1.131	Derived	Least-squares fit
r_DM μ (combo), σ (combo)	7.113, 2.480	Derived	Ratio fit
r_DM μ (pembro), σ (pembro)	7.042, 5.000	Derived	Ratio fit
r_LR μ (combo), σ (combo)	8.874, 5.000	Derived	Ratio fit
r_LR μ (pembro), σ (pembro)	5.913, 2.674	Derived	Ratio fit
Long-term extrapolation			
10-year OS, combo	84.1%	Modeled	Log-normal extrapolation
10-year OS, pembro	51.6%	Modeled	Log-normal extrapolation
20-year OS, combo	60.1%	Modeled	GP hazard floor
(GP-constrained)			
20-year OS, pembro	28.5%	Modeled	GP hazard floor
(GP-constrained)			
30-year OS, combo	22.2%	Modeled	GP hazard floor
(GP-constrained)			
30-year OS, pembro	10.5%	Modeled	GP hazard floor
(GP-constrained)			
40-year OS, combo	2.0%	Modeled	GP hazard floor
(GP-constrained)			

3.1 Observed vs. Fitted Survival Curves



Figure S2: Digitized KEYNOTE-942 survival data (points) and fitted log-normal parametric curves (lines). Left panel: intismeran plus pembrolizumab arm. Right panel: pembrolizumab alone arm.

4 Survival Model Selection

4.1 Component-Level Parametric Fit

Table S4: Component-Level Parametric Fit Across Five Distributions

Component	Distribution	AIC	RMSE	Rank
<i>OS – Combo arm</i>				
Exponential	1 param	-45.49	0.0087	4
Weibull	2 param	-49.84	0.0046	2
Log-normal	2 param	-50.08	0.0045	1
Log-logistic	2 param	-49.89	0.0046	3
Generalized gamma	3 param	-47.97	0.0045	5
<i>OS – Pembro arm</i>				
Exponential	1 param	-31.27	0.0359	4
Weibull	2 param	-32.00	0.0273	1
Log-normal	2 param	-31.16	0.0297	3
Log-logistic	2 param	-31.69	0.0282	2
Generalized gamma	3 param	-30.33	0.0264	5
<i>r_{DM} – Combo arm</i>				
Exponential	1 param	-36.79	0.0078	3
Weibull	2 param	-36.53	0.0063	4
Log-normal	2 param	-36.93	0.0060	1
Log-logistic	2 param	-36.61	0.0062	2
Generalized gamma	3 param	-34.74	0.0061	5
<i>r_{DM} – Pembro arm</i>				
Exponential	1 param	-19.41	0.0688	5
Weibull	2 param	-28.43	0.0174	3
Log-normal	2 param	-27.23	0.0202	4
Log-logistic	2 param	-30.30	0.0137	2
Generalized gamma	3 param	-32.45	0.0082	1
<i>r_{LR} – Combo arm</i>				
Exponential	1 param	-30.38	0.0392	5
Weibull	2 param	-38.97	0.0136	3
Log-normal	2 param	-39.49	0.0129	2
Log-logistic	2 param	-39.70	0.0126	1
Generalized gamma	3 param	-38.43	0.0118	4
<i>r_{LR} – Pembro arm</i>				
Exponential	1 param	-31.82	0.0340	5
Weibull	2 param	-33.60	0.0233	1
Log-normal	2 param	-33.37	0.0238	2
Log-logistic	2 param	-33.51	0.0235	3
Generalized gamma	3 param	-31.69	0.0231	4

4.2 Joint Model Selection

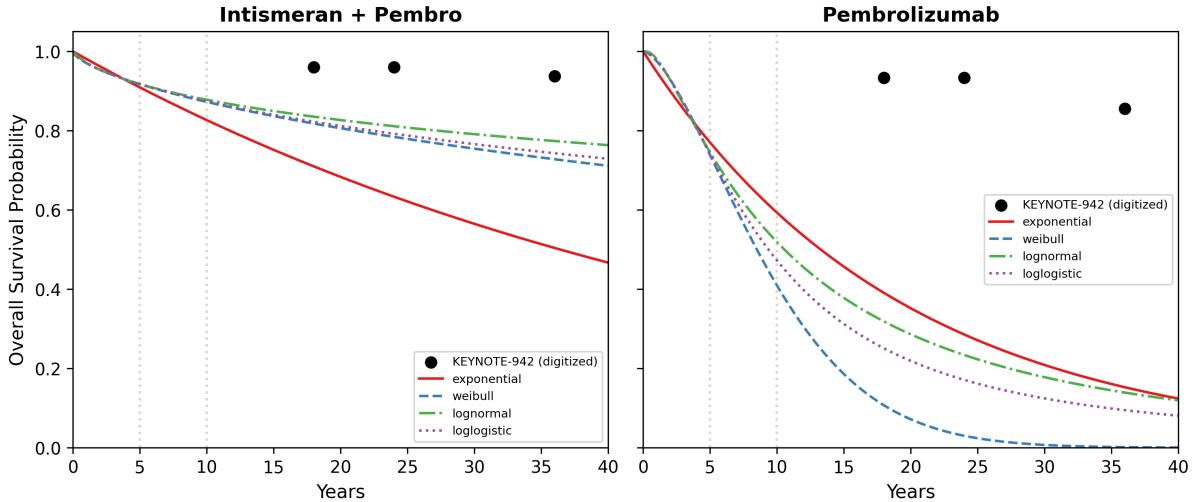
A total of 15,625 distribution combinations (5 distributions x 6 curves) were evaluated for structural validity at 40 years. All 15,625 satisfied the $RFS \leq DMFS \leq OS$ ordering constraint (0 crossing violations). The full results are available online ([output/survival_combos_15625.csv](#)).

Table S5: Top 10 Joint Model Combinations by Total AIC (5 distributions, 15,625 combinations)

Rank	OS	r _{DM}	r _{LR}	Total AIC	OS40c	OS40p
	C/P	C/P	C/P			
1	LN / W	LN / GG	LL / W	-224.76	76.3%	0.0%
2	LN / W	LN / GG	LL / LL	-224.67	76.3%	0.0%
3	LN / W	Exp / GG	LL / W	-224.62	76.3%	0.0%
4	LLog / W	LN / GG	LL / W	-224.57	72.9%	0.0%
5	LN / W	LN / GG	LN / W	-224.55	76.3%	0.0%
6	LN / W	LN / GG	LL / LN	-224.53	76.3%	0.0%
7	LN / W	Exp / GG	LL / LL	-224.53	76.3%	0.0%
8	W / W	LN / GG	LL / W	-224.52	71.1%	0.0%
9	LLog / W	LN / GG	LL / LL	-224.48	72.9%	0.0%
10	LN / W	LN / GG	LN / LL	-224.46	76.3%	0.0%

Distributions: LN = log-normal, W = weibull, LLog = log-logistic, Exp = exponential, GG = generalized gamma. C/P = Combo/Pembro. All top-ranked combinations use log-normal OS for the combo arm, yielding OS40c = 71.1–76.3% under unconstrained extrapolation, substantially higher than GP survival at age 101 (2.0%). This motivated the GP mortality hazard floor (Section 5.1).

4.3 Lifetime OS Extrapolation by Distribution



Vertical dotted lines at 5 and 10 years mark the boundary between within-trial and extrapolated periods.

Figure S3: Lifetime OS extrapolation by parametric distribution. Within-trial data (0–5 years) are shown to the left of the first vertical dotted line; beyond 10 years is pure extrapolation. The four conventional distributions produce wildly divergent 40-year survival estimates (0% to 76.3%). Generalized gamma was included in the formal model-selection analysis but is not shown here for visual clarity.

5 General-Population Mortality and Structural Validation

5.1 GP Mortality Implementation

The GP mortality hazard floor is applied to modeled OS from month 60 onward.

Life table source: US 2019 period life tables (Arias et al., NVSS). **Baseline age:** 61 years (KEYNOTE-942 median). **Sex weighting:** 64% male, 36% female. **Interpolation:** Annual life

table survival linearly interpolated to monthly resolution; monthly hazards derived as $h_{GP}(t) = -\ln[S_{GP}(t+1)/S_{GP}(t)]$. **Constraint formula (mortality hazard floor):** from month 60 onward, the model mortality hazard is floored at the age- and sex-matched general-population hazard:

$$h_{\text{model}}^{\text{con}}(t) = \max[h_{\text{param}}(t), h_{GP}(t)] \quad t \geq 60 \text{ months}, \quad (1)$$

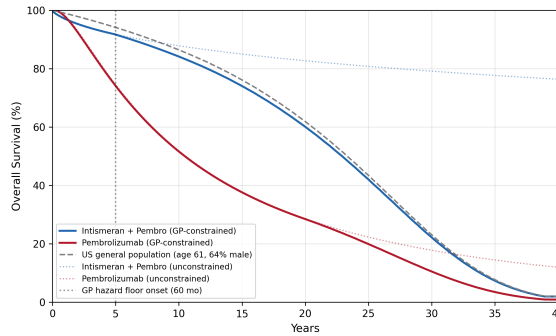
where $h_{\text{param}}(t) = -\ln[S_{\text{param}}(t+1)/S_{\text{param}}(t)]$ is the monthly hazard implied by the fitted parametric survival curve. The constrained survival is reconstructed as $S_{\text{model}}^{\text{con}}(t) = \exp[-\sum_0^t h^{\text{con}}(u)]$. This is a **hazard floor**, not a survival cap: a cohort that has already experienced elevated (cancer-related) mortality in the first 5 years remains below general-population survival at every subsequent time point, whereas a survival cap ($S = \min[S_{\text{param}}, S_{GP}]$) would incorrectly force the two curves to coincide whenever the parametric tail exceeded the general-population level. Both arms are subject to the same background mortality floor.

5.2 Long-Term Survival Diagnostics

Table S6: Long-Term Survival Predictions vs. General Population. GP survival is conditional on being alive at baseline age 61 and is calculated as $S_{GP}(t) = l_{61+t}/l_{61}$ using the sex-weighted 2019 US life tables (64% male, 36% female); the terminal 100+ life-table value is used for the 40-year row.

Years from baseline	Attained age	Combo OS	Pembro OS	GP survival
5	66	91.6%	74.1%	94.1%
10	71	84.1%	51.6%	86.4%
15	76	74.0%	37.6%	76.1%
20	81	60.1%	28.5%	61.9%
30	91	22.2%	10.5%	23.1%
40	101	2.0%	0.9%	2.0%

5.3 OS versus General-Population Survival



GP-constrained curves (solid) use a mortality hazard floor from month 60. The constrained combo OS stays below the GP curve at all time points, reflecting the excess mortality experienced in the first 5 years. Unconstrained log-normal (dotted) produces implausibly high 40-year survival (76.4%).

Figure S4: OS versus general population. Solid lines: GP-constrained model (hazard floor from 60 months). Dotted lines: unconstrained log-normal extrapolation. Dashed gray line: US GP survival from age 61. The constrained combo curve remains below the GP curve at all time points, reflecting the hazard-floor property: a cohort with prior excess mortality does not "catch up" to general-population survival.

5.4 Structural Quality Assurance

Table S7: Structural Quality Assurance (480 Monthly Cycles)

Check	Result
Total cycles	480
RFS > DMFS violations	0
DMFS > OS violations	0
Negative RF state probability	0
Negative LR state probability	0
Negative DM state probability	0
Maximum state-sum error	$< 10^{-12}$
Clamp events (forced non-negative)	0

6 External Plausibility Assessment

6.1 External Plausibility Assessment Landmarks

Table S8: External Plausibility Assessment of Pembrolizumab Arm Predictions

Outcome	Year	Model (pembro)	KEYNOTE-054	CheckMate 238
RFS	5	40.5%	—	51% [6]
RFS	7	31.9%	50% [7]	46% [6]
DMFS	5	53.8%	—	59% [6]
DMFS	7	44.9%	54% [7]	55% [6]
OS	5	74.5%	NR	76% [8]
OS	7	64.2%	NR	71% [6]

NR: Not reported. CheckMate 238 evaluated nivolumab (not pembrolizumab). The model predicts lower RFS/DMFS versus external trials, reflecting the higher-risk KEYNOTE-942 population (stage IIIB–IV including resected stage IV melanoma vs IIIA–C in KEYNOTE-054 and IIIB–IV in CheckMate 238). Only published landmark values are included; no extrapolated estimates are used.[6, 7]

6.2 Longitudinal External Plausibility Assessment

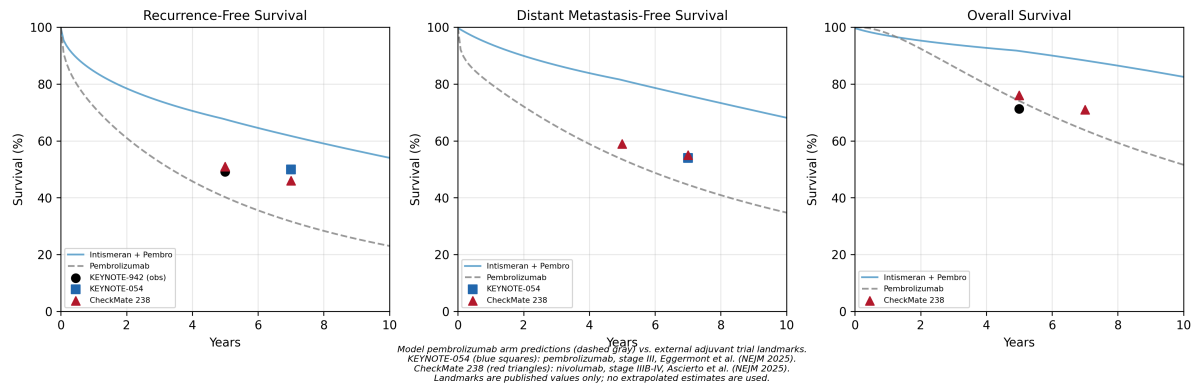


Figure S5: External plausibility assessment of pembrolizumab arm predictions. Gray line: model. Circles: KEYNOTE-942 observed data. Squares: KEYNOTE-054 (pembrolizumab, 7-year). Triangles: CheckMate 238 (nivolumab, 5- and 7-year). Only published landmark values are shown.

7 Model Inputs

7.1 Complete Parameter Table

Table S9: Full Model Parameter Inventory. *Note:* DSA ranges are based on fixed percentage perturbations ($\pm 20\%$ or $\pm 40\%$) of the base-case fitted values to explore parameter sensitivity. PSA distributions use Normal sampling with standard errors derived from the nonlinear least-squares Jacobian covariance, reflecting curve-fit uncertainty rather than the full sampling uncertainty of the original censored trial data.

Parameter	Base	DSA Low	DSA High	PSA Distribution	Source
<i>Clinical Parameters</i>					
OS μ – combo	8.396	5.038 (-40%)	11.754 ($+40\%$)	Normal (SE from Jacobian)	Fitted
OS σ – combo	3.096	1.858 (-40%)	4.334 ($+40\%$)	Normal (SE from Jacobian)	Fitted
OS μ – pembro	4.841	3.873 (-20%)	5.809 ($+20\%$)	Normal (SE from Jacobian)	Fitted
OS σ – pembro	1.131	0.905 (-20%)	1.357 ($+20\%$)	Normal (SE from Jacobian)	Fitted
r _{DM} μ – combo	7.113	4.268 (-40%)	9.958 ($+40\%$)	Normal (SE from Jacobian)	Fitted
r _{DM} σ – combo	2.480	1.488 (-40%)	3.472 ($+40\%$)	Normal (SE from Jacobian)	Fitted
r _{DM} μ – pembro	7.042	5.634 (-20%)	8.450 ($+20\%$)	Normal (SE from Jacobian)	Fitted
r _{DM} σ – pembro	5.000	3.000 (-40%)	7.000 ($+40\%$)	Normal (SE from Jacobian)	Fitted
r _{LR} μ – combo	8.874	5.324 (-40%)	12.424 ($+40\%$)	Normal (SE from Jacobian)	Fitted
r _{LR} σ – combo	5.000	3.000 (-40%)	7.000 ($+40\%$)	Normal (SE from Jacobian)	Fitted
r _{LR} μ – pembro	5.913	4.730 (-20%)	7.096 ($+20\%$)	Normal (SE from Jacobian)	Fitted
r _{LR} σ – pembro	2.674	1.604 (-40%)	3.744 ($+40\%$)	Normal (SE from Jacobian)	Fitted
Treatment duration (pembro)	18 cycles (2 yr)	9	24	Fixed	[4]
Treatment duration (intismeran)	9 doses (1 yr)	6	12	Fixed	[4]
Baseline age	61 years	55	67	Fixed	[4]
Male proportion	64%	–	–	Fixed	[4]
<i>Costs (2026 USD)</i>					
Intismeran (one-time)	\$200,000	\$160,000	\$240,000	Fixed (not sampled)	Assumed
Pembrolizumab (annual)	\$220,896	\$176,717	\$265,075	Gamma (CV=0.2)	WAC (CMS)
Tumor sequencing	\$1,000	\$500	\$2,000	Gamma (CV=0.2)	CMS
Administration (per cycle)	\$500	\$250	\$750	Gamma (CV=0.2)	CMS
AE management (one-time)	\$15,000	\$7,500	\$22,500	Gamma (CV=0.2)	[3]
LR management (monthly)	\$3,000	\$2,400	\$3,600	Gamma (CV=0.2)	[3]
DM management (monthly)	\$12,000	\$9,600	\$14,400	Gamma (CV=0.2)	[1]
<i>Utilities</i>					
RF state	0.83	0.80	0.86	Beta (SE=0.03)	[2]
LR state	0.64	0.61	0.67	Beta (SE=0.03)	[2]
DM state	0.55	0.52	0.58	Beta (SE=0.03)	[2]
AE disutility (one-time)	0.05	0.00	0.10	Beta (SE=0.02)	Assumed
<i>Economic Parameters</i>					
Discount rate (costs, effects)	3%	0%	5%	Fixed (DSA only)	[9]
Time horizon	40 years	10	20	Fixed (scenario)	Assumed
Willingness-to-pay threshold	\$100K–\$150K/QALY	\$50K	\$200K	Fixed (scenario)	[10]
Cost base year	2026 USD	–	–	–	Assumed
GP mortality source	US 2019 Life Tables	–	–	–	NVSS (Arias 2022)

8 Additional Results

8.1 Lifetime Cost Decomposition

Table S10: Lifetime Cost Decomposition by Component (GP-constrained, 40-year horizon)

Component	Combo	Pembro	Incremental
Intismeran	\$200,000	\$0	\$200,000
Pembrolizumab	\$217,888	\$217,888	\$0
Sequencing	\$1,000	\$0	\$1,000
Administration	\$493	\$493	\$0
Adverse events	\$15,000	\$0	\$15,000
LR management	\$85,455	\$69,341	\$16,114
DM management	\$354,189	\$424,388	\$-70,199
Total	\$874,026	\$712,110	\$161,916

8.2 LY and QALY Decomposition

Table S11: Life-Year and QALY Decomposition by Health State

Outcome	Combo	Pembro	Incremental
Life-years (undiscounted)	21.58	13.81	7.77
Life-years (discounted 3%)	15.06	10.28	4.78
RF state QALYs	8.49	4.48	4.01
LR state QALYs	1.52	1.23	0.29
DM state QALYs	1.35	1.62	-0.27
Total QALYs	11.35	7.33	4.02

Health benefit is driven by RF state QALYs (4.01 gained), with a modest additional contribution from LR (+0.29) and a small reduction in DM state QALYs (−0.27) under the corrected mortality-constrained extrapolation.

9 Uncertainty and Structural Scenario Analyses

9.1 Deterministic Sensitivity Analysis

Table S12: One-Way Deterministic Sensitivity Analysis

Parameter	Base	Low	High	ICER range	Change in NMB
Utility RF	0.83	0.80	0.86	41,777–38,875	\$43,396
Utility LR	0.64	0.61	0.67	40,409–40,140	\$4,029
Utility DM	0.55	0.52	0.58	40,128–40,421	\$4,387
Pembrolizumab annual cost	\$220,896	\$176,717	\$265,075	40,274–40,274	\$0
Intismeran cost	\$200,000	\$160,000	\$240,000	30,324–50,223	\$80,000
LR monthly cost	\$3,000	\$2,400	\$3,600	39,472–41,075	\$6,446
DM monthly cost	\$12,000	\$9,600	\$14,400	43,766–36,782	\$28,079
Discount rate	3%	0%	5%	33,120–48,284	\$434,030
OS μ combo	8.396	-40%	+40%	19,875–40,684	\$485,457
OS μ pembro	4.841	-20%	+20%	57,434–Dominant	\$441,968

The maximum finite ICER was \$57,434/QALY across the tested ranges; the combination became dominant when the pembrolizumab OS parameter was increased by 20%.

9.2 Probabilistic Sensitivity Analysis

9.2.1 PSA Implementation Details

PSA with 6,000 Monte Carlo iterations (seed = 42). Survival parameters (μ , σ for 6 curves) drawn jointly from bivariate normal via Jacobian covariance ($r > 0.95$); σ lower-truncated at 0.05. Costs: gamma (CV=0.2). Utilities: beta (SE=0.03). Structural constraints ($RFS \leq DMFS \leq OS$, monotonicity, survival within $[0,1]$) verified every cycle; all 6,000 draws were structurally valid (0 rejections). No iterations were excluded on the basis of the ICER value.

Table S13: PSA Parameter Distributions and Sampled Values

Parameter	Distribution	Fitted mean	SE	Correlation	Sampling
OS μ / σ – combo	Bivariate normal	8.396 / 3.096	0.793 / 0.522	0.996	Joint
OS μ / σ – pembro	Bivariate normal	4.841 / 1.131	0.291 / 0.310	0.955	Joint
r _{DM} μ / σ – combo	Bivariate normal	7.113 / 2.480	0.814 / 0.563	0.995	Joint
r _{DM} μ / σ – pembro	Bivariate normal	7.042 / 5.000	2.278 / 3.130	0.995	Joint
r _{LR} μ / σ – combo	Bivariate normal	8.874 / 5.000	2.039 / 1.928	0.997	Joint
r _{LR} μ / σ – pembro	Bivariate normal	5.913 / 2.674	0.757 / 0.879	0.984	Joint
Intismeran cost	Fixed	\$200,000	—	—	Fixed (not sampled)
Pembrolizumab annual cost	Gamma (CV=0.2)	\$220,896	—	—	Independent
LR monthly cost	Gamma (CV=0.2)	\$3,000	—	—	Independent
DM monthly cost	Gamma (CV=0.2)	\$12,000	—	—	Independent
Utility RF / LR / DM	Beta (SE=0.03)	0.83 / 0.64 / 0.55	—	—	Independent

9.2.2 PSA Results

- Mean Δ QALY: 4.32 (95% CI 2.53–7.04).
- Mean Δ Cost: \$127,711 (95% CI –\$668,394 to \$520,672).
- Mean NMB: \$304,291 at \$100K and \$520,293 at \$150K.
- Probability cost-effective: 63.8% at \$50K, 95.3% at \$100K, and 98.7% at \$150K.
- No iterations were excluded based on the ICER; 16.7% of draws were cost-saving, and all 6,000 draws had Δ QALY > 0.

9.2.3 PSA Convergence

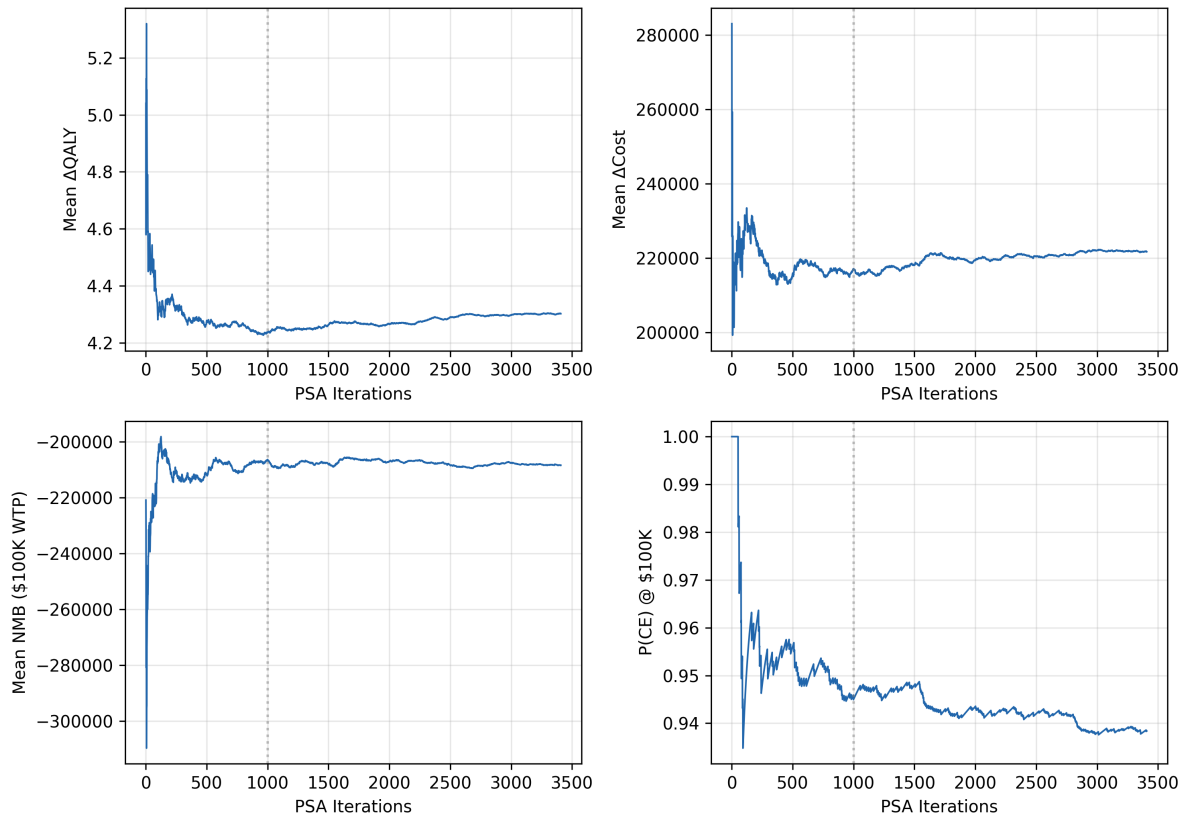


Figure S6: PSA convergence across iteration counts (6,000 structurally valid iterations, 0 structural rejections, seed = 42). Cumulative means of Δ QALY, Δ Cost, NMB, and P(CE). Vertical dotted lines at 500, 1,000, 2,000, and 4,000 iterations. Estimates stable from ~500 iterations.

9.2.4 Cost-Effectiveness Plane

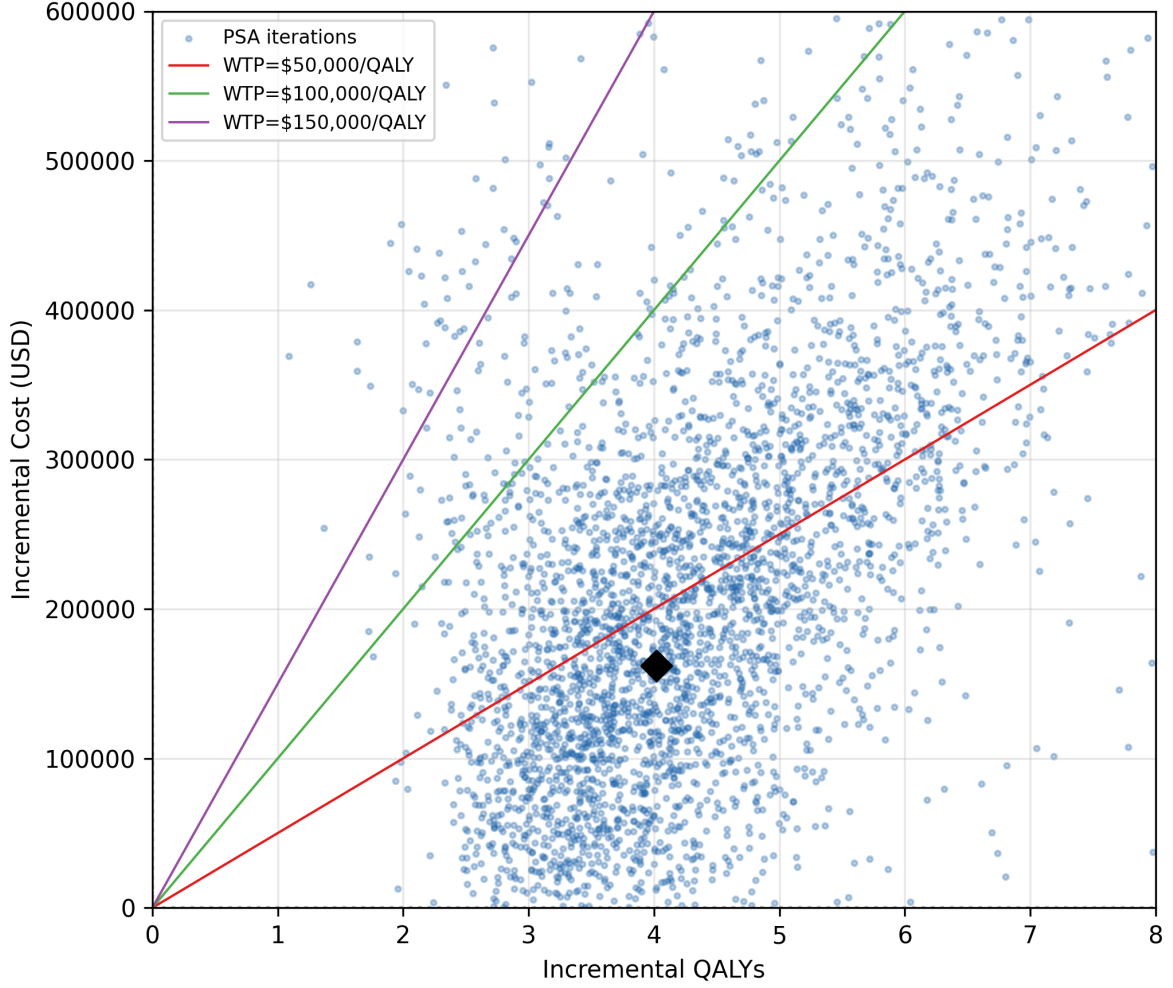


Figure S7: CE plane showing 6,000 PSA iterations (blue), WTP lines (\$50K, \$100K, \$150K per QALY), and the deterministic base case (black diamond). 16.7% of iterations fall in the southeast quadrant (cost-saving); all 6,000 iterations had positive Δ QALY. No iterations were excluded based on the ICER value.

9.3 Structural Scenario Analyses

Scenario implementations. Three structural scenarios alter the base-case treatment-effect assumptions. All scenarios preserve the observed 5-year KEYNOTE-942 data unchanged and modify only the extrapolated trajectory beyond 60 months.

OS-survival convergence (5–20 y, Waning V1). On the **survival probability scale**, the incremental OS benefit (combo OS minus pembrolizumab OS) is linearly attenuated from its full base-case value at 60 months to zero at 240 months (20 years). At each monthly cycle t : $S_{\text{combo}}^{\text{waned}}(t) = S_{\text{pembro}}(t) + [S_{\text{combo}}^{\text{base}}(t) - S_{\text{pembro}}(t)] \times \max(0, 1 - \frac{t-60}{180})$. The recurrence (r_{DM}) and local-recurrence (r_{LR}) ratio functions are **not** modified, so the scenario isolates the effect of attenuating the OS benefit while preserving the observed recurrence advantage. The general-population hazard floor continues to apply independently to both arms.

Hazard convergence (5–10 y, Waning V2). On the **cause-specific mortality hazard scale**, the combo arm's monthly all-cause hazard is linearly blended toward the pembrolizumab arm's hazard from month 60 to month 120 (10 years). At each cycle: $h_{\text{combo}}^{\text{waned}}(t) = (1 - w(t))h_{\text{combo}}(t) + w(t)h_{\text{pembro}}(t)$, where $w(t) = \max(0, \min(\frac{t-60}{60}, 1))$. The waned hazard is integrated forward to reconstruct the survival curve: $S_{\text{combo}}^{\text{waned}}(t+1) = S_{\text{combo}}^{\text{waned}}(t) \times \exp(-h_{\text{combo}}^{\text{waned}}(t))$, with $S_{\text{combo}}^{\text{waned}}(t) = S_{\text{combo}}^{\text{base}}(t)$ for $t \leq 60$ months. In this

scenario, r_{DM} and r_{LR} are also linearly converged to pembrolizumab-arm values on the probability scale ($\text{HR} \rightarrow 1$ by 10 years), so that all three survival components reflect complete treatment-effect erosion. The GP hazard floor continues to apply.

No direct OS benefit. The combination arm's OS curve is set equal to the pembrolizumab arm's OS curve at all time points ($S_{\text{combo}}(t) = S_{\text{pembro}}(t) \forall t$). *Therecurrence*(r_{DM}) and local-recurrence (r_{LR}) ratio functions retain their base-case values, so any incremental QALY gain arises solely from the improved disease-state trajectory (longer recurrence-free and distant-metastasis-free intervals) rather than from extended overall survival. $\text{RFS} \leq \text{DMFS} \leq \text{OS}$ ordering is maintained trivially since both arms share the same OS curve.

Table S14: Structural Scenario Analysis Results

Scenario	ΔLY	$\Delta QALY$	$\Delta Cost$	ICER	NMB \$100K	NMB \$150K
Base case	4.78	4.02	\$161,916	\$40,274	\$240,123	\$441,143
Weibull OS	5.60	5.60	\$307,578	\$54,925	\$252,422	\$532,422
Unconstrained (no GP)	9.24	7.22	\$373,488	\$51,753	\$348,512	\$709,512
GP floor 48 mo	4.78	4.00	\$160,954	\$40,236	\$239,046	\$439,046
GP floor 60 mo (base)	4.78	4.02	\$161,916	\$40,274	\$240,123	\$441,143
GP floor 72 mo	4.78	4.05	\$163,611	\$40,351	\$241,389	\$443,889
GP floor 96 mo	4.91	4.15	\$168,745	\$40,623	\$246,255	\$453,755
10-year horizon	1.30	1.30	\$96,400	\$74,305	\$33,600	\$98,600
20-year horizon	3.54	3.08	\$123,974	\$40,293	\$183,705	\$337,545
30-year horizon	3.89	3.89	\$155,242	\$39,914	\$233,758	\$428,258
40-year horizon (base)	4.78	4.02	\$161,916	\$40,274	\$240,123	\$441,143
Intismeran \$100K	4.78	4.02	\$61,916	\$15,400	\$340,123	\$541,143
Intismeran \$200K (base)	4.78	4.02	\$161,916	\$40,274	\$240,123	\$441,143
Intismeran \$300K	4.78	4.02	\$261,916	\$65,147	\$140,123	\$341,143
Intismeran \$500K	4.78	4.02	\$461,916	\$114,893	-\$59,877	\$141,143
Waning V1 (OS-survival convergence)	1.91	1.93	\$41,571	\$21,514	\$151,429	\$248,929
Waning V2 (hazard convergence)	2.72	2.17	\$227,542	\$104,852	-\$10,530	\$97,975
No direct OS benefit	0.00	0.50	-\$14,330	Dominant	\$64,330	\$89,330

The base-case conclusion is robust across structural assumptions. The ICER alone is insufficient: unconstrained model has moderate ICER (\$51,753) but implausible \$373,488 cost.

Time horizon sensitivity. The ICER remained below conventional US willingness-to-pay thresholds across 20-, 30-, and 40-year horizons, although the point estimate increased at the 40-year horizon: \$40,293 at 20 years, \$39,914 at 30 years, and \$40,274 at the 40-year base-case horizon (Table S14). The 30-to-40-year increase of \$360/QALY (0.9%) reflects the limited contribution of late discounted QALY gains beyond 30 years; the 10-year horizon produced a higher ICER of \$74,305/QALY, reflecting the incomplete capture of long-term QALY gains within a shorter window. Because the general-population mortality hazard floor was applied from 60 months onward and both costs and QALYs were discounted at 3% annually, survival beyond the 20-year window contributed materially to discounted incremental costs and QALYs.

General population hazard floor start-time sensitivity. The GP mortality floor was applied from 60 months in the base case. Varying the onset from 48 to 96 months produced no material change in the ICER (the ICERs ranged from \$40,236 to \$40,623/QALY across 48–96 months). This reflects the fact that the fitted hazard for the combination arm closely tracks the general-population mortality hazard in the post-trial period (both at approximately 0.0008–0.0009 monthly), leaving the floor with minimal additional constraining effect regardless of onset timing. The floor’s binding is thus a near-numerical coincidence between the parametric extrapolation and US life-table mortality, not a model artifact.

10 Deterministic Threshold Acquisition Price

10.1 Price Threshold Analysis

Table S15: Price Threshold Analysis

Intismeran Price	ICER	NMB @ \$100K	NMB @ \$150K
\$0	Dominant	\$440,123	\$641,143
\$50,000	\$2,964	\$390,123	\$591,143
\$100,000	\$15,400	\$340,123	\$541,143
\$150,000	\$27,837	\$290,123	\$491,143
\$200,000 (base)	\$40,274	\$240,123	\$441,143
\$250,000	\$52,710	\$190,123	\$391,143
\$300,000	\$65,147	\$140,123	\$341,143
\$350,000	\$77,583	\$90,123	\$291,143
\$400,000	\$90,020	\$40,123	\$241,143
\$450,000	\$102,457	-\$9,877	\$191,143
\$500,000	\$114,893	-\$59,877	\$141,143
\$440,063	\$100,000	\$0	\$200,000
\$550,000	\$127,330	-\$109,877	\$91,143
\$600,000	\$139,766	-\$159,877	\$41,143
\$650,000	\$152,203	-\$209,877	-\$8,857
\$700,000	\$164,640	-\$259,877	-\$58,857
\$750,000	\$177,076	-\$309,877	-\$108,857
\$800,000	\$189,513	-\$359,877	-\$158,857
\$640,869	\$150,000	-\$200,746	\$0

Deterministic threshold prices (from Table S15): approximately \$440,063 at \$100,000/QALY (2.2× base) and approximately \$640,869 at \$150,000/QALY (3.2× base). These are model-based deterministic estimates; PSA uncertainty propagation and formal budget-impact analysis would be needed to inform payer negotiations.

11 Reproducibility and Reporting

11.1 Technical Reproducibility

All analyses were performed using Python 3.12.7 (Apple Clang 15.0.0) on macOS 26.6.2 (Apple Silicon). The full code repository is available at <https://github.com/yichao2022/cea-intismeran> (tag v1.0-submission).

Key dependencies: NumPy 1.26.4, SciPy 1.14.1, Matplotlib 3.9.3, Pandas 2.2.3, LaTeX (TeX Live 2025, pdfLaTeX + BibTeX + natbib).

Model details:

- Cycle length: monthly (1/12 year); time horizon: 40 years (480 cycles)
- Half-cycle correction: not applied (monthly cycles are sufficiently fine-grained)
- Discounting: annual discount factor $(1 + r)^{-t}$ with t measured in years, where $r = 0.03$
- Random seed: 42 (PSA reproducibility)
- GP mortality source: US 2019 period life tables (NVSS, Arias et al. 2022)

Code structure:

- `model.py` — ModelParams, CEAModel (4-state PSM)
- `rerun_primary.py` — CEAModelV2 (hazard-based GP floor, waning, no-OS)
- `psa_full.py` — PSA with joint survival sampling
- `survival_model_selection.py` — 15,625-combination model selection
- `final_run.py` / `threshold_analysis.py` — DSA, scenarios, pricing
- `figures.py` / `gen_fig_*.py` — Figure generation scripts
- `output/survival_combos_15625.csv` — Full model selection results

Data extraction: Survival probabilities extracted from Figure 1 of Khattak et al. (2026) using WebPlotDigitizer v4.7. The digitized values are in Table S2.

Reproducibility: Running `python3 rerun_primary.py` reproduces base-case results. `python3 psa_full.py` reproduces PSA. GitHub tag v1.0-submission freezes the submission code.

11.2 CHEERS 2022 Checklist

Table S16: CHEERS 2022 Checklist

No.	Item	Reported	Location
1	Title	Yes	Title page
2	Abstract	Yes	Abstract
3	Background	Yes	Introduction
4	Health economic analysis plan	No	This is a retrospective analysis, not a pre-specified HEAP
5	Model	Yes	Methods, Appendix A
6	Population	Yes	Methods
7	Setting	Yes	Methods
8	Comparators	Yes	Methods
9	Perspective	Yes	Methods
10	Time horizon	Yes	Methods
11	Discount rate	Yes	Methods
12	Outcomes	Yes	Methods
13	Health states	Yes	Appendix A
14	Measurement of outcomes	Yes	Methods
15	Resource use	Yes	Methods
16	Currency	Yes	Methods
17	Model choice	Yes	Methods
18	Assumptions	Yes	Methods (§2.2–2.7)
19	Analytic methods	Yes	Methods (§2.7)
20	Study parameters	Yes	Methods (§2.4, 2.5), Appendix D
21	Incremental results	Yes	Results (§3.1–3.2)
22	Uncertainty	Yes	Results (§3.4–3.6), Supplement
23	Heterogeneity	Not as- sessed	Limitations (§4.7, subgroup heterogeneity acknowledged)
24	Discussion	Yes	Discussion
25	Limitations	Yes	Discussion
26	Funding	Yes	Declarations
27	Conflicts	Yes	Declarations
28	Reproducibility	Yes	Appendix

References

- [1] A. G. Bensimon, Z. Y. Zhou, M. Jenkins, et al. Cost-effectiveness of pembrolizumab for the adjuvant treatment of resected high-risk stage iii melanoma in the united states. *Journal of Medical Economics*, 22(10):981–993, 2019.
- [2] K. M. Johnston et al. Health-state utility values in melanoma. *Pharmacoeconomics*, 39:241–253, 2021.
- [3] S. Zhang, A. G. Bensimon, R. Xu, et al. Cost-effectiveness analysis of pembrolizumab as an adjuvant treatment of resected stage iib or iic melanoma in the united states. *Advances in Therapy*, 40:3038–3055, 2023.
- [4] A. Khattak, M. S. Carlino, T. Meniawy, G. Ansstas, M. H. Taylor, K. B. Kim, et al. Intismeran autogene plus pembrolizumab versus pembrolizumab alone in high-risk resected melanoma: 5-year update of the randomized phase 2b keynote-942 study. *Journal of Clinical Oncology*, 44:JCO2600835, 2026. doi: 10.1200/JCO-26-00835.
- [5] J. S. Weber, M. S. Carlino, A. Khattak, et al. Individualised neoantigen therapy mrna-4157 (v940) plus pembrolizumab versus pembrolizumab monotherapy in resected melanoma (keynote-942): a randomised, phase 2b study. *The Lancet*, 403(10421):632–644, 2024.
- [6] P. A. Ascierto, M. Del Vecchio, B. Merelli, et al. Nivolumab for resected stage iii or iv melanoma at 9 years. *New England Journal of Medicine*, 394(4):333–342, 2026. doi: 10.1056/NEJMoa2504966.
- [7] A. M. M. Eggermont, M. Kicinski, C. U. Blank, et al. Seven-year analysis of adjuvant pembrolizumab versus placebo in stage iii melanoma in the eortc1325 / keynote-054 trial. *European Journal of Cancer*, 211:114327, 2024. doi: 10.1016/j.ejca.2024.114327.
- [8] P. A. Ascierto, M. Del Vecchio, M. Mandalá, et al. Adjuvant nivolumab versus ipilimumab in resected stage iii/iv melanoma: 5-year efficacy and biomarker results from checkmate 238. *Clinical Cancer Research*, 29(17):3352–3361, 2023.
- [9] G. D. Sanders, P. J. Neumann, A. Basu, et al. Recommendations for conduct, methodological practices, and reporting of cost-effectiveness analyses: Second panel on cost-effectiveness in health and medicine. *JAMA*, 316(10):1093–1103, 2016.
- [10] P. J. Neumann, J. T. Cohen, and M. C. Weinstein. Updating cost-effectiveness — the curious resilience of the \$50,000-per-qaly threshold. *New England Journal of Medicine*, 371(9):796–797, 2014.